Cheng Chen

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RESEARCH SUMMARY

3D Computer Vision and Autonomous Driving — My research focuses on semantic scene understanding, spanning collaborative 3D semantic occupancy perception, BEV map segmentation, lane topology reasoning, and 4D world modeling for forecasting and planning. I develop structured scene representations and efficient inference methods using Gaussian-based modeling, generative priors, and knowledge distillation. My current work explores distilling autoregressive models into feedforward perception networks and improving the accuracy and efficiency of lane topology prediction. My broader interests include multi-modal perception, open-world generalization through foundation models, and applying large language models (LLMs), vision-language models (VLMs), and vision-language-action models (VLAs) to perception, reasoning, and planning for autonomous driving.

Technical Skills —
Programming Languages: Python (NumPy, OpenCV, Pandas), C/C++, Java, Shell script, MATLAB, HTML/JS
Deep/Machine Learning: PyTorch (3D, Video), Scikit-learn (metrics, decomposition), Einops, Transformers, SpConv, LangChain

EDUCATION

Purdue University, West Lafayette, IN, USA — Aug 2022 – Dec 2027
Ph.D. in Computer Engineering — GPA: 4.0/4.0
Research area: *Computer Vision Systems*, supervised by Dr. Saurabh Bagchi.

New York University, Brooklyn, NY, USA — Sept 2020 – May 2022
M.Sc. in Computer Engineering — GPA: 4.0/4.0

University of California, Berkeley, Berkeley, CA, USA — Jan 2019 – May 2019
Exchange Student in Electrical Engineering and Computer Science — GPA: 3.9/4.0

Nanjing University, Nanjing, Jiangsu, China — Sept 2016 – June 2020
B.Sc. in Computer Science (*Elite Program*) — GPA: 4.38/5.0 Rank: 5/29

PUBLICATIONS

- **Cheng Chen**, et al. GEM: Gaussian Evolution Model for Occupancy Forecasting and Motion Planning. Manuscript submitted.
- **Cheng Chen**, et al. Vision-Only Gaussian Splatting for Collaborative Semantic Occupancy Prediction. In *the Proceedings of the 40th AAAI Conference on Artificial Intelligence* (**AAAI 2026 Oral**).
- **Cheng Chen**, et al. E2D2: Energy-Efficient and Deadline-Driven Ensemble Inference at the Edge. Manuscript submitted.
- Yuan-Yao Lou, **Cheng Chen**, et al. AD-VRAN: DRL-based Adaptive Deployment of Virtualized RAN in an Open Telco Edge Cloud. In *IEEE Journal on Selected Areas in Communications* (**JSAC 2026**).
- Hao Huang, **Cheng Chen**, Yi Fang. Manifold Adversarial Learning for Cross-domain 3D Shape Representation. In *the 17th European Conference on Computer Vision* (**ECCV 2022**).
- Hao Huang, **Cheng Chen**, Yi Fang. Non-Rigid Multiple Point Set Registration Using Latent Gaussian Mixture. In *the 29th IEEE International Conference on Image Processing* (**ICIP 2022**).

EXPERIENCE

Perception PhD Intern, *Waymo* — May 2026 – Aug 2026

Mentors: Xianghao Xu, Kyle Zhang, Shiyu Song

- Proposed a camera-only BEV segmentation framework that transfers autoregressive map priors through expanding-horizon feature distillation, achieving state-of-the-art performance on nuScenes and Argoverse 2 while running at over 11 FPS.

Research Assistant, *Dependable Computing Systems Laboratory*, Purdue University — Aug 2022 – Present

Mentors: Dr. Saurabh Bagchi, Dr. Kwang Taik Kim, Dr. Hao Huang

- **Evolving Gaussians for 4D Occupancy Forecasting and Motion Planning:**
  - Proposed the first continuous occupancy world model that represents dynamic scenes using interpretable evolving 4D Gaussian primitives, enabling arbitrary timestamp querying without sequential decoding over fixed future steps.
  - Developed a unified end-to-end framework that combines primitive-based scene evolution with ego-motion reasoning.

- Achieved state-of-the-art performance in both 4D occupancy forecasting and motion planning.

- **Sparse Gaussians for collaborative 3D semantic occupancy:**
  - Proposed the first vision-only framework using sparse 3D semantic Gaussian splatting for multi-vehicle perception.
  - Designed neighborhood-based cross-agent attention to fuse Gaussian primitives, removing duplicates and suppressing noise.
  - Achieved mIoU gains of **+8.42** (vs. single-agent) and **+3.28** (vs. collaborative), and IoU gains of **+5.11** and **+22.41**.

- **Deadline-aware collaborative inference on the Edge:**
  - Introduced the first deadline-driven, energy-efficient ensemble inference framework for heterogeneous edge devices.
  - Demonstrated ensemble accuracy gains across diverse models, modalities, and data distributions.
  - Built a six-device NVIDIA Jetson testbed with ZeroMQ and message quantization, emulating dynamic real-world conditions.
  - Achieved up to **+4.06%** accuracy improvement and over **60%** reduction in energy waste under strict deadline guarantees.

- **Deep reinforcement learning (DRL) for next-generation communication:**
  - Formulated a DRL-based policy for optimized deployment of virtualized Radio Access Networks (vRAN).
  - Achieved up to **17%** reduction in resource consumption while improving QoS.
  - Supported emerging Beyond-5G (B5G) and 6G applications requiring adaptive and efficient networking.

Research Assistant, *Multimedia and Visual Computing Lab*, New York University — May 2021 – May 2022

Mentors: Dr. Fang Yi, Dr. Hao Huang

- **Adversarial learning for Sim2Real:**
  - Proposed a 3D point cloud representation method for domain generalization.
  - Unified geometric transformations within a Lie group manifold using geodesic distance.
  - Integrated adaptive memory to retain effective adversarial samples, improving robustness in transfer.

- **Weakly supervised indoor scene segmentation:**
  - Developed a low-rank sparse point transformer for large-scale point cloud segmentation.
  - Achieved strong performance using only **0.1%** labeled points, reducing annotation cost.

TEACHING, MENTORSHIP & SERVICE

Teaching Assistant: *ECE 20001 Electrical Engineering Fundamentals*, Purdue University — Summer 2025
- Led recitation sessions on core problem sets and held office hours to address students' questions.
- Managed coordination among TAs to respond efficiently to ad-hoc tutoring requests.

Teaching Assistant: *ECE Senior Design*, Purdue University — 2025 & 2026
- Mentored four senior design teams, providing structured feedback on journals and assignments.
- Supported all teams with PCB verification, software debugging, and overall project guidance.

Teaching Assistant: *CS-GY 6953 / ECE-GY 7123 Deep Learning*, New York University — Spring 2022
- Designed mini-projects and developed auto-grading scripts to streamline evaluation.

Undergraduate Research Mentor, Purdue University — Spring 2024
- *Measurement-based Experiments and Evaluation for Digital Transformation in Smart Factory*. Mentored Ana Milena Espinosa Jiménez through the Undergraduate Research Experience Purdue-Colombia (UREP-C) Program.
- *Latency, Velocity, and Accuracy Evaluation of Classification Models on Edge Devices*. Mentored Ana Milena Espinosa Jiménez under the Summer Undergraduate Research Fellowship (SURF).

Reviewer: *ICLR, KDD, AAAI, NeurIPS, JSAC, ICIP*

SELECTED AWARDS

Phi Kappa Phi Honor Society (**Top 10%**), Purdue University — 2025

Best M.S. Academic Achievement (**Top 3**), New York University — 2022

Graduate School of Engineering Scholarship, New York University — 2020

International and Domestic Exchange Award, Nanjing University — 2019

National Elite Program Scholarship, Nanjing University — 2018